Improving Group Assignment Submission and Mark Capture

A Process Improvement Case Study

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Identifying repeated work, testing an existing platform capability, and introducing a reusable departmental process.

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1 How the Improvement Started

When I raised this approach, the Finance teaching team I was working with was not aware that Amathuba could support the group submission and marking workflow we needed. Other courses, particularly in Computer Science, were already using it.

Having seen the approach used elsewhere, I researched how it worked and whether it could address the difficulties we faced with group assignments in Finance. I then proposed it to the department.

The idea was straightforward: students complete the assignment as a group, so the platform should recognise that group when receiving the submission and recording the result. One student could submit on behalf of the group, and a marker could capture one shared mark rather than entering the same result separately for every member.

I received approval to test the approach in FTX1005F. During the trial, we retained the normal process as a backup. This allowed us to check the new workflow while still having a familiar way to complete the assessment if it did not work as expected.

I tested it with two colleagues, and it worked well. The experience gave us a practical basis for discussing wider use within the department.

2 The Original Process

2.1 Group Work, Individual Mark Capture

The investment project was completed in groups. A marker evaluated the group's submission and arrived at one shared project mark. However, recording that result still required individual entries for the students in the group.

For each submission, the marker had to:

1. identify the students belonging to the group;
2. assess the group's work;
3. determine the shared mark;
4. find each member's individual record;
5. enter the same mark for every member; and
6. check that the individual entries were consistent.

For a group of five students, the work was assessed once, but the resulting mark had to be captured five times.

For example, if a group received 72 out of 100, the intended records were:

Group member	Project mark
Student A	72
Student B	72
Student C	72
Student D	72
Student E	72

These are fictional records used to illustrate the process. The five entries represent the same assessment decision.

2.2 Disadvantages of the Manual Approach

Repeated administrative work. The marker had to record a result for every person, even though the students had completed the same group assignment. The number of capture actions therefore grew with the number of students rather than the number of submissions being evaluated.

Inconsistent marks within a group. Repeated entry created opportunities to type a different mark for one member, select the wrong student record, or miss a member entirely. For example, entering 72 for four students and 27 for the fifth would create different results from the same group assessment.

This is an illustration of the risk, rather than a measured error from the trial. An individual difference may be valid where the course approves an adjustment, but it should be deliberate and documented.

Additional checking and correction. Markers and lecturers had to check that each student received the intended mark. If an entry was missing or inconsistent, they needed to trace the student's group and correct the record.

More effort when a group mark changed. If a group's result was revised after moderation, every individual entry needed to be updated consistently. This repeated the same administrative task and created another opportunity for a mismatch.

Membership administration. The teaching team needed a reliable account of who belonged to each group. Submission records, membership information and individual grade entries had to remain aligned.

The concern was therefore larger than the time taken to type a mark. It was the effort needed to keep several records of one shared result consistent.

3 The Proposed Workflow

I proposed using Amathuba's existing group-assignment capability to connect the students, their submission and the shared evaluation.

The revised workflow was:

1. record students in their project groups on Amathuba;
2. confirm membership before the assignment began;

3. keep the recorded membership fixed during the assignment, with necessary changes handled by the teaching team;
4. allow one member to submit for the group;
5. assess the group's submission once; and
6. capture one shared evaluation through the group assignment so the result applies to its members.

The academic judgement remained with the marker. The improvement concerned submission, membership management and mark capture.

3.1 Why Use an Existing Platform Capability?

The course already used Amathuba. The capability was also being used in other courses, which provided a practical starting point for my research.

Using the existing platform meant that the department did not need another application, a separate database, or a custom integration to address this particular problem.

My contribution was to recognise a useful practice elsewhere, understand it, propose it in the Finance context, and help test whether it worked for our assessment process.

4 Approval and Testing

4.1 A Trial With the Existing Process as Backup

After receiving approval, I tested the workflow in FTX1005F with two colleagues. We kept the normal process available as a backup while trying the new method.

This mattered because the change affected assessed work. The trial needed to establish that the workflow was useful without leaving the teaching team dependent on an unfamiliar process before we had confidence in it.

During the trial, retaining the backup also meant that we should not treat the trial itself as the full time-saving result. Running or maintaining two approaches can add work. The efficiency calculations later in this report describe the revised workflow when used as the normal process.

4.2 What the Trial Established

The workflow worked well in practice, and the colleagues who tested it found the process easier to manage. It showed that the department could use group submission and shared mark capture for this assessment.

The positive experience supported the next conversation: whether the approach should continue and be adapted for other Finance courses.

This report does not claim a measured reduction in marking errors or a timed before-and-after study. Those would require records of errors and elapsed time from comparable marking rounds.

5 Who Benefited

5.1 Students

For students, the process provided a clear way to submit group work: one member could upload the final assignment on behalf of everyone in the recorded group.

Keeping membership fixed once the assignment began also addressed the concern that someone could leave or change the recorded group midway through the work. Necessary changes could still be considered by the teaching team, rather than being made informally during the assignment.

Stable membership depended on the way the process was configured and managed. Selecting a group assignment alone should not be described as automatically preventing every membership change.

5.2 Markers

Markers could focus on assessing each submission and recording its shared result once. They no longer needed to repeat the same capture for each group member.

This reduced the manual work involved in distributing the mark and removed opportunities for differences caused by repeated typing.

5.3 Lecturers and Course Coordinators

For lecturers, the workflow made the relationship between the group, its submission and its mark clearer. It reduced the amount of individual capture that needed to be checked and the repeated updates required when a shared result changed.

Lecturers still needed to oversee assessment quality, membership exceptions, moderation and release. The improvement reduced repeated administration around those responsibilities.

6 Calculating the Reduction in Mark Capture

6.1 Assumptions

The estimates below use:

- approximately 150 students in FTX1005F;
- approximately 300 students in each of FTX3044 and FTX3045;
- an average of five students per group; and
- one shared mark per group, before any approved individual adjustments.

The class sizes are approximate. The calculations therefore illustrate the expected scale of the improvement rather than an audited count of the actual groups.

The calculations for FTX3044 and FTX3045 are presented per course, assuming approximately 300 students in each. They do not establish that both courses had already implemented the workflow.

6.2 Calculation Method

Let:

$$N = \text{number of students}, \qquad s = \text{average group size}.$$

The estimated number of groups is:

$$G = \frac{N}{s}.$$

Under individual capture, the number of manual entries is:

$$C_{\text{old}} = N.$$

Under shared group capture:

$$C_{\text{new}} = G.$$

The reduction in manual entries is:

$$\Delta C = N - G.$$

The percentage reduction is:

$$\text{Capture reduction} = \frac{N - G}{N} \times 100\%.$$

6.3 FTX1005F: Approximately 150 Students

With an average group size of five:

$$G = \frac{150}{5} = 30 \text{ groups}.$$

The old process required approximately 150 mark captures. The revised process requires approximately 30 shared captures:

$$\Delta C = 150 - 30 = 120.$$

Therefore:

$$\text{Capture reduction} = \frac{120}{150} \times 100\% = \boxed{80\%}.$$

The capture-count ratio is:

$$\frac{150}{30} = \boxed{5:1}.$$

This means the revised workflow requires one-fifth as many manual mark entries for the shared results.

6.4 A Course With Approximately 300 Students

For either FTX3044 or FTX3045, using the stated estimate of 300 students:

$$G = \frac{300}{5} = 60 \text{ groups}.$$

The reduction becomes:

$$\Delta C = 300 - 60 = 240.$$

Therefore:

$$\text{Capture reduction} = \frac{240}{300} \times 100\% = \boxed{80\%}.$$

The capture-count ratio remains:

$$\frac{300}{60} = \boxed{5:1}.$$

The percentage improvement is the same, but the larger course removes twice as many repeated entries as FTX1005F.

6.5 Summary

Course	Students	Groups	Old captures	New captures
FTX1005F	150	30	150	30
FTX3044	300	60	300	60
FTX3045	300	60	300	60
Combined scenario	750	150	750	150

If all three courses used the workflow under these assumptions:

$$750 - 150 = \boxed{\text{600 fewer manual entries}}$$

per comparable assessment round.

This combined figure is a potential rollout scenario, not a claim that all three courses had already used it.

7 What the Calculations Mean for Time

An 80% reduction in mark entries does not mean that the entire assessment takes 80% less time. Markers still need to read the work, apply the rubric, write feedback and complete moderation.

The direct saving concerns repeated mark capture.

If one individual capture takes an average of t seconds, and a shared group capture takes the same time, the estimated capture time saved is:

$$T_{\text{saved}} = (N - G)t.$$

To express this in minutes:

$$T_{\text{saved, minutes}} = \frac{(N - G)t}{60}.$$

For example, using an **illustrative, unmeasured** capture time of 30 seconds:

Course or scenario	**Old time**	**New time**	**Saving**
FTX1005F	75 min	15 min	60 min
FTX3044	150 min	30 min	120 min
FTX3045	150 min	30 min	120 min
Combined scenario	375 min	75 min	300 min

Under that illustrative assumption, the combined saving would be five hours of capture time per assessment round.

These are scenario calculations, not measured hours saved. Actual results depend on the time needed to locate records, enter marks, configure groups, check membership and resolve exceptions.

A fuller comparison would use:

$$T_{\text{old capture}} = Nt_i,$$

$$T_{\text{new capture}} = Gt_g,$$

where t_i is the average individual capture time and t_g is the average shared group capture time.

Net administrative time saved would then account for any additional setup and verification work:

$$T_{\text{net saved}} = Nt_i - Gt_g - T_{\text{additional administration}}.$$

The defensible numerical result from the class-size assumptions is therefore **80% fewer manual shared-mark entries**, rather than a measured fivefold improvement in total marking speed.

8 Departmental Response and Wider Use

After the successful trial, the Finance department wanted to continue with the approach and adapt it for other courses. I documented the process so that another teaching team could understand the setup and repeat it.

The earlier project records describe adoption in FTX3044F and planned use in FTX4056S. FTX3045 is included in this report's calculations because of its stated class size; its implementation status is not established here.

The wider value was that the department could reuse a process it had tested rather than treating the improvement as something that depended on me configuring it each time.

Each course still needed to check its own assessment rules, membership arrangements and individual exceptions. The approach was suitable where a submission received a shared group mark.

9 My Contribution and the Skills Demonstrated

This project required me to connect an operational problem with a capability that was already available.

Skill	How I applied it
Problem identification	I recognised repeated capture and the difficulties of keeping group members' marks consistent.
Research and initiative	I investigated an approach used in other courses and proposed it within Finance.
User-centred thinking	I considered student submission and membership alongside the needs of markers and lecturers.
Technical judgement	I used an existing Amathuba capability rather than introducing another system.
Testing and risk management	I tested with colleagues while retaining the normal process as a backup.
Stakeholder communication	I sought approval, shared the trial experience and supported discussion of wider use.
Documentation and handover	I wrote down the process so other teaching teams could repeat it.
Analytical reasoning	I distinguished the reduction in capture actions from unmeasured claims about total time saved.

10 Reflection

What I liked about this project was that the problem did not require a large new system. It required understanding where people were doing repeated work and looking at how other courses had already approached it.

The improvement also needed to work for more than one person. Students needed a clear way to submit together and a stable recorded group. Markers needed to capture the shared result efficiently. Lecturers needed a process they could oversee and continue using.

Researching the capability was only the starting point. Getting approval, testing with colleagues, keeping a backup and documenting the result made the proposal practical enough for the department to consider adopting.

If I repeated the work, I would measure capture time before and after the change and retain a clear validation record from the beginning. That would allow me to show the measured time saving alongside the reduction in manual entries and the feedback from the teaching team.